

IT-Simplera Institute



Week 1

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Networking Fundamentals & Network Simulation Tools

Introduction

Networking is the process of connecting devices so they can share data and resources. It is a core part of modern IT systems and forms the backbone of most organizations.

This report presents the work completed in Week 1 of the Network Administration Internship Program at IT-Simplera Institute. The focus of this week was to build a solid understanding of basic networking concepts. This included network devices, IP addressing, and communication models.

To gain practical experience, two widely used simulation tools were studied. These were Cisco Packet Tracer and GNS3. Both tools are used by network professionals to design and test networks in a virtual environment before applying them on real hardware.

A basic LAN topology was built using both tools. Devices were assigned static IP addresses. The ping command was then used to test and confirm connectivity between them. This report explains the steps followed, the concepts understood, and the difficulties faced during the task.

Networking Concepts Learned

Network: A network is a group of devices connected together to share data and resources.

LAN (Local Area Network): A LAN connects devices within a small area, such as an office or a lab. It is used for fast and local communication.

IP Address: An IP address is a unique number assigned to each device on a network. It allows devices to identify and communicate with each other.

Static IP Addressing: In static addressing, an IP address is manually assigned to a device. It does not change unless updated manually.

Network Devices:

- Router: Connects different networks and directs data between them.
- Switch: Connects multiple devices within the same network and forwards data to the correct device.
- PC / End Device: Sends and receives data on the network.

Communication Models: Networking follows standard models like OSI and TCP/IP. These models divide network communication into layers. Each layer has a specific role in sending and receiving data.

Ping Command: Ping is used to test connectivity between two devices. It sends small packets of data and checks if a response is received.

Cisco Packet Tracer Interface Overview

Cisco Packet Tracer is a network simulation tool developed by Cisco. It is used to design, build, and test network topologies in a virtual environment.

Main Parts of the Interface:

Device Panel: Located at the bottom left. It contains categories of network devices like routers, switches, and end devices.

Workspace: The large area in the center. Devices are dragged here to build the topology.

Connections Panel: Contains different types of cables used to connect devices, such as copper straight-through and console cables.

Physical/Logical Toggle: Allows switching between a logical view of the network and a physical view.

Simulation Mode: Used to observe how data packets travel through the network step by step.

Command Line Interface (CLI): Available inside routers and switches. Used to configure devices using IOS commands.

Cisco Packet Tracer — LAN Topology

A simple Local Area Network was built in Cisco Packet Tracer. The topology consisted of one Router (2911), one Switch (2960), two PCs (PC0 and PC1), and one Laptop. All devices were connected using copper straight-through cables. The router interface was configured with a static IP address and served as the default gateway for all end devices.

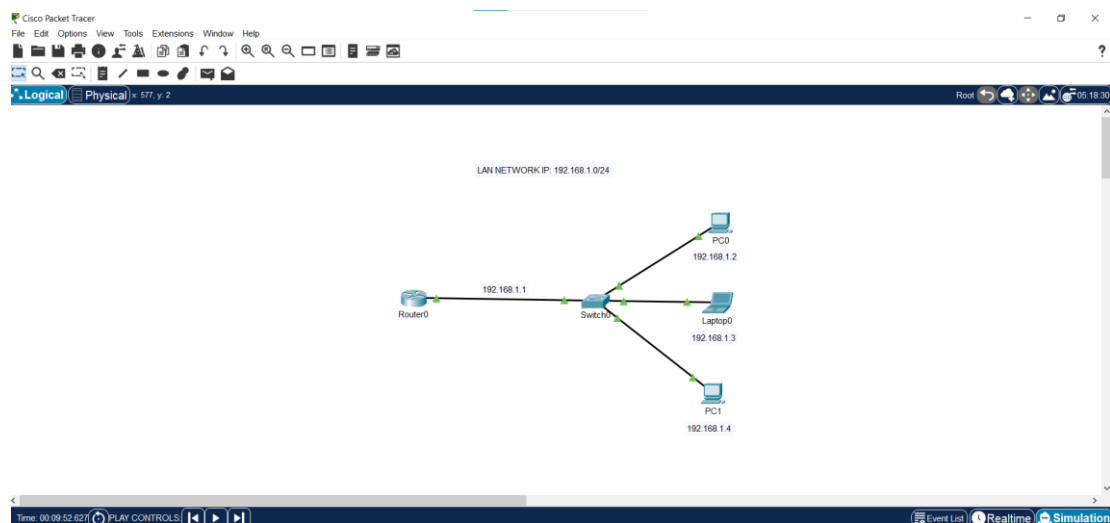


Figure 1: LAN Topology in Cisco Packet Tracer — Network: 192.168.1.0/24

IP Address Configuration

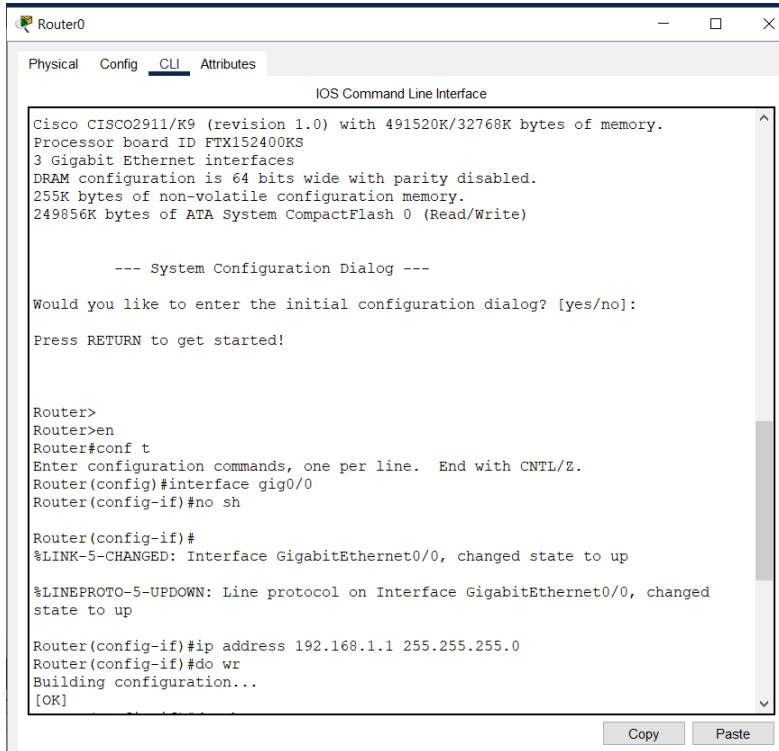
The following static IP addresses were manually assigned to all devices in the network:

Device	Interface	IP Address	Subnet Mask	Gateway
Router0	GigabitEthernet0/0	192.168.1.1	255.255.255.0	—
PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
Laptop0	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.4	255.255.255.0	192.168.1.1

Router Configuration (CLI)

The router's GigabitEthernet0/0 interface was configured using the CLI. The interface was enabled using the 'no shutdown' command, the IP address 192.168.1.1 was assigned with a subnet mask of 255.255.255.0, and the configuration was saved using 'do wr'.

Router0 — CLI Configuration:



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!

Router>
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

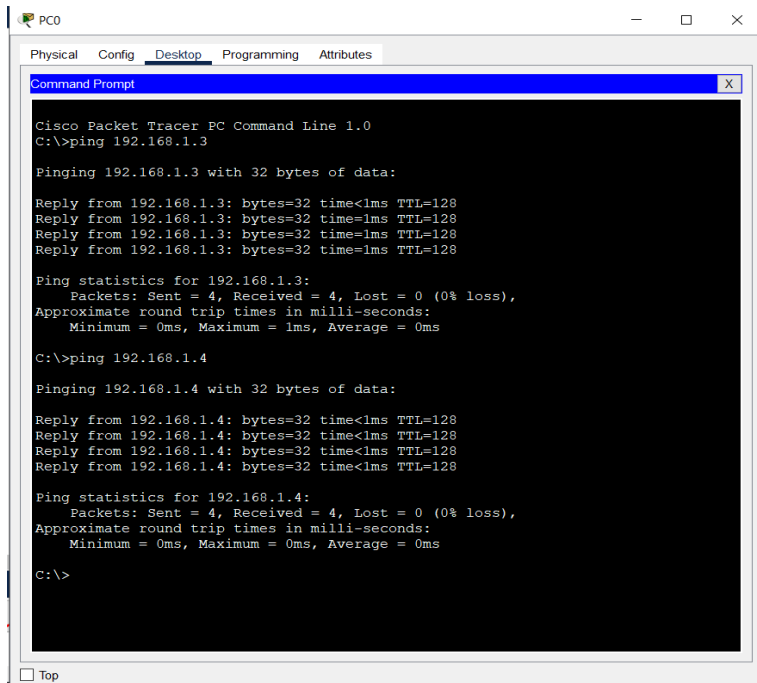
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed
state to up

Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#do wr
Building configuration...
[OK]
```

Ping Test Results — Cisco Packet Tracer

After all IP addresses were configured, connectivity was tested using the ping command from each device. Each device pinged the other two devices on the network.

PC0 — IP: 192.168.1.2 | Pinging Laptop0 (192.168.1.3) and PC1 (192.168.1.4):



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.4

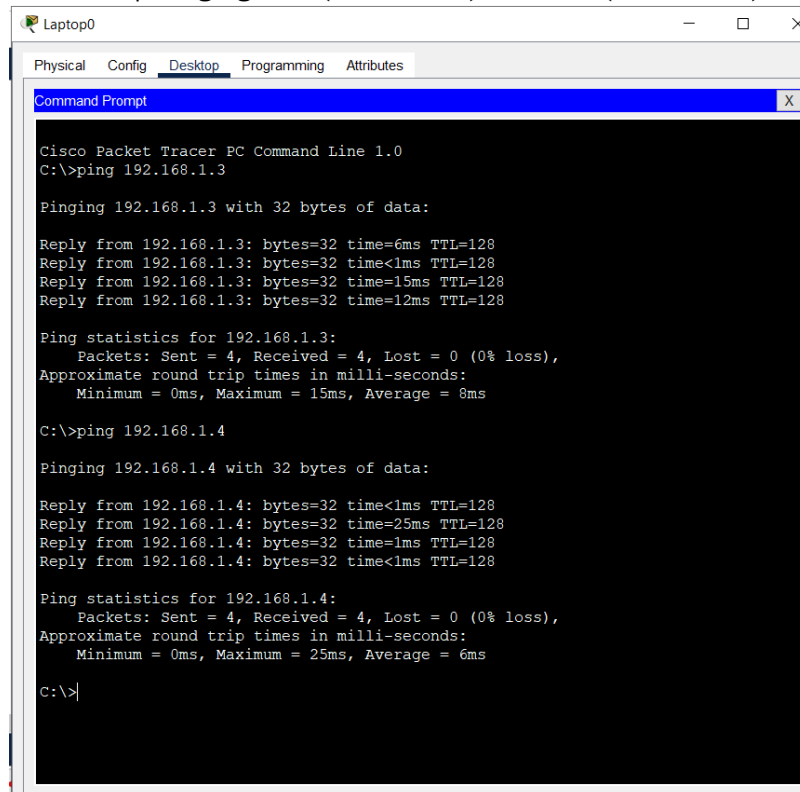
Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Laptop0 — IP: 192.168.1.3 | Pinging PC0 (192.168.1.2) and PC1 (192.168.1.4):



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=6ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=15ms TTL=128
Reply from 192.168.1.3: bytes=32 time=12ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 15ms, Average = 8ms

C:\>ping 192.168.1.4

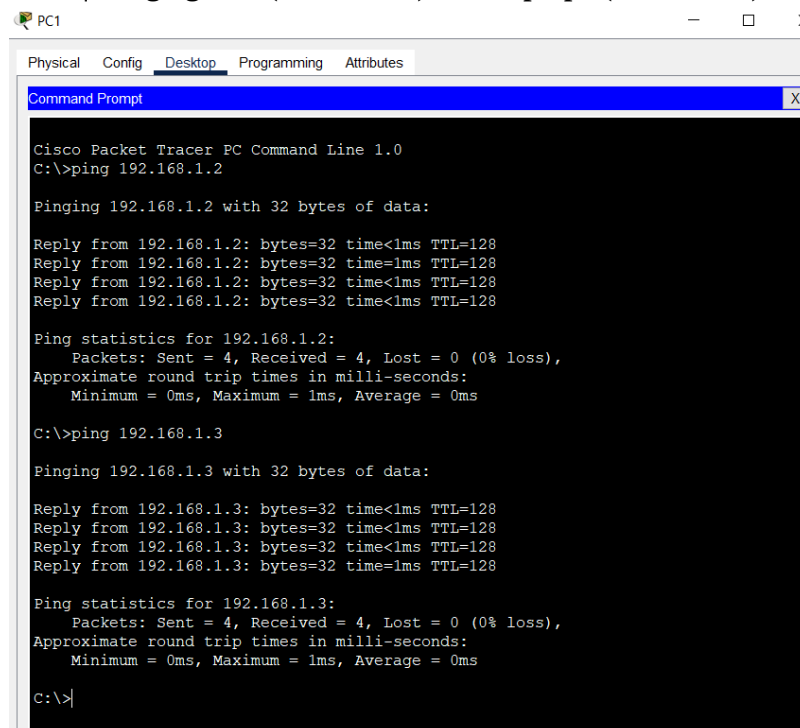
Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=25ms TTL=128
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 25ms, Average = 6ms

C:\>
```

PC1 — IP: 192.168.1.4 | Pinging PC0 (192.168.1.2) and Laptop0 (192.168.1.3):



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Result: All ping tests in Cisco Packet Tracer returned 4/4 replies with 0% packet loss. The LAN was working correctly.

GNS3 Interface Overview

GNS3 (Graphical Network Simulator-3) is a network simulation tool used to design and test network configurations using virtual and emulated devices. Unlike Cisco Packet Tracer which simulates devices, GNS3 runs actual network operating system images, making it much closer to real-world networking.

Main Parts of the Interface:

Devices Toolbar: Located on the left side. Contains routers, switches, and VPCS end devices that can be added to the project.

Workspace: The central area where devices are placed and connected to build the topology.

Topology Summary: Right-side panel showing all nodes in the project along with their console ports and running status.

Console Access: Each device is accessed through a terminal. VPCS nodes open directly; routers use PuTTY or the built-in terminal.

Server Summary: Shows the GNS3 local server status and system information.

Start/Stop Controls: Toolbar buttons to start, pause, or stop all devices in the project simultaneously.

GNS3 — LAN Topology

The same LAN topology was built again in GNS3 using VPCS (Virtual PC Simulator) nodes and an Ethernet Switch. Three VPCS nodes — PC1, PC2, and PC3 — were connected to Switch1. Static IP addresses were assigned to each VPCS node through the terminal console.

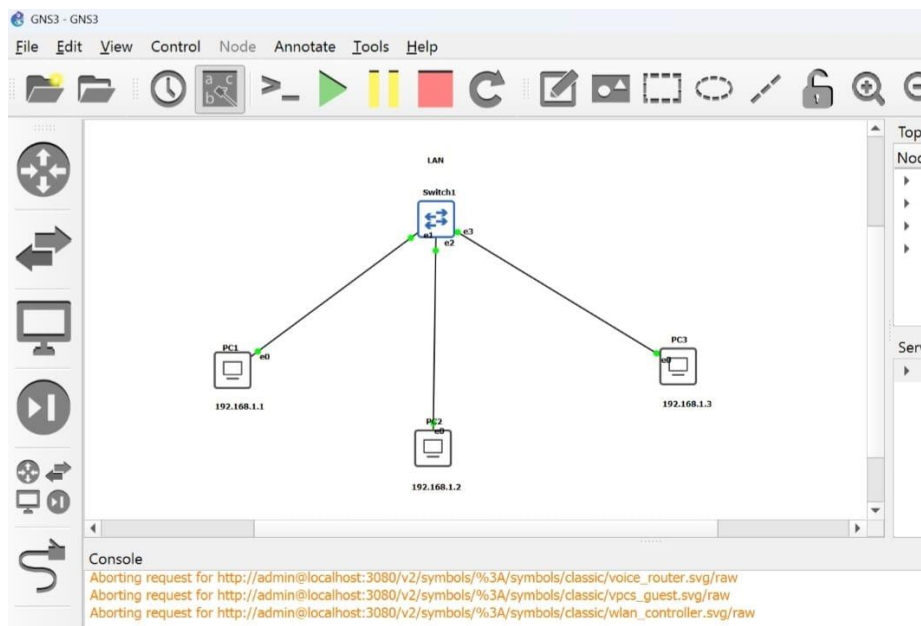


Figure 2: LAN Topology in GNS3 — Switch1 with PC1 (192.168.1.1), PC2 (192.168.1.2), PC3 (192.168.1.3)

IP Address Configuration — GNS3

IP addresses were assigned to each VPCS node using the `ip` command in their terminal consoles. After assigning, configurations were saved using the `save` command.

Device	Command Used	IP Address	Subnet Mask
PC1 (VPCS)	<code>ip 192.168.1.1 255.255.255.0</code>	192.168.1.1	255.255.255.0
PC2 (VPCS)	<code>ip 192.168.1.2 255.255.255.0</code>	192.168.1.2	255.255.255.0

Device	Command Used	IP Address	Subnet Mask
PC3 (VPCS)	ip 192.168.1.3 255.255.255.0	192.168.1.3	255.255.255.0

Ping Test Results — GNS3

After configuring all three VPCS nodes, ping tests were run from each device to verify connectivity across the LAN. Each device pinged the other two nodes on the network.

PC1 — IP: 192.168.1.1 | Pinging PC2 (192.168.1.2) and PC3 (192.168.1.3):

```

PC1> ip 192.168.1.1 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0

PC1> ping 192.168.1.2
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=0.888 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=1.032 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=1.352 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=0.813 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=0.871 ms

PC1> ping 192.168.1.3
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=1.169 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=0.968 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=1.292 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=1.549 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=1.311 ms

PC1> save
Saving startup configuration to startup.vpc
. done
PC1>

```

PC2 — IP: 192.168.1.2 | Pinging PC1 (192.168.1.1) and PC3 (192.168.1.3):

```

PC2> ip 192.168.1.2 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.2 255.255.255.0

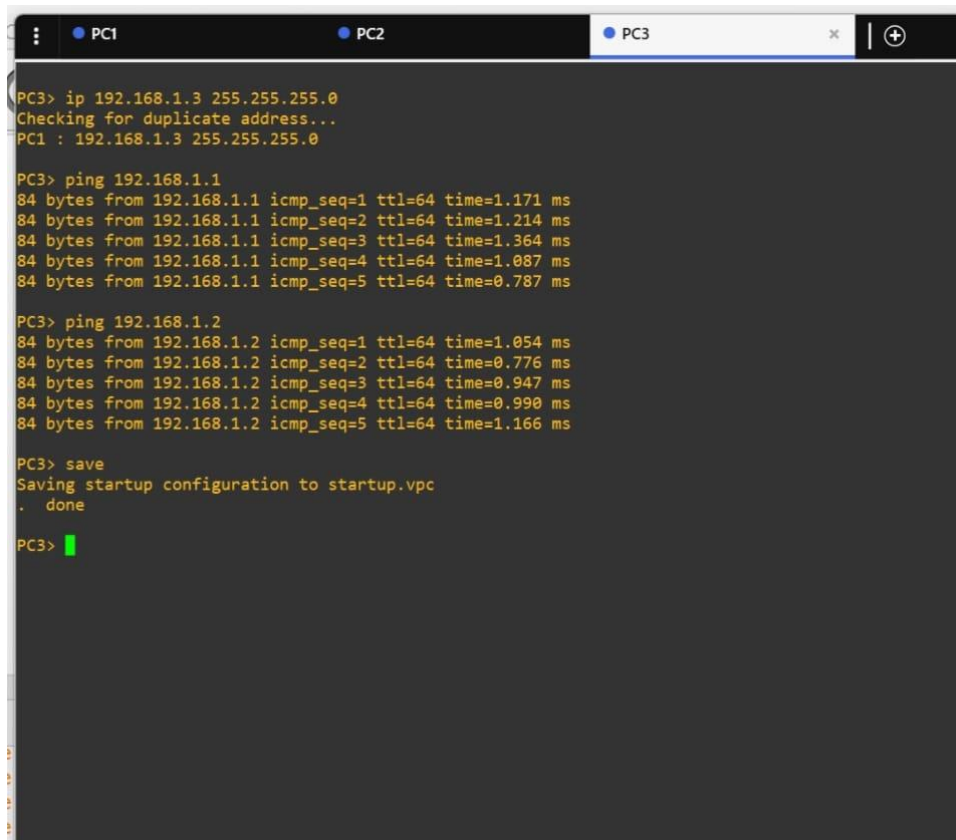
PC2> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=1.009 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=1.340 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=1.037 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=0.833 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=0.885 ms

PC2> ping 192.168.1.3
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=1.574 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=1.040 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=0.775 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=1.173 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=1.051 ms

PC2> save
Saving startup configuration to startup.vpc
. done
PC2>

```

PC3 — IP: 192.168.1.3 | Pinging PC1 (192.168.1.1) and PC2 (192.168.1.2):

A screenshot of a GNS3 console window for PC3. The window has a dark background with yellow text. At the top, there are tabs for PC1, PC2, and PC3, with PC3 being the active tab. The console output shows the configuration of IP 192.168.1.3 on interface 255.255.255.0, followed by two ping commands. The first ping is to 192.168.1.1, showing five successful replies with varying times. The second ping is to 192.168.1.2, also showing five successful replies. Finally, the 'save' command is executed, and the configuration is saved to startup.vpc.

```
PC3> ip 192.168.1.3 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.3 255.255.255.0

PC3> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=1.171 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=1.214 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=1.364 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=1.087 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=0.787 ms

PC3> ping 192.168.1.2
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=1.054 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=0.776 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=0.947 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=0.990 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=1.166 ms

PC3> save
Saving startup configuration to startup.vpc
. done

PC3>
```

Result: All GNS3 ping tests returned 5/5 replies with 0% packet loss. Full connectivity was confirmed across all VPCS nodes.

Challenges Faced

I was already familiar with Cisco Packet Tracer from previous coursework and lab work. Because of this, I did not face any major difficulty while building the topology or configuring the devices in Packet Tracer.

GNS3 was completely new to me. I had never used it before this task. I started by exploring and understanding its interface, since it is quite different from Packet Tracer. The following difficulties were encountered:

- **Installation Issues** — GNS3 requires a background server process running on port 3080. On initial launch the server failed to start and GNS3 displayed a connection timeout error. This required troubleshooting the server startup process.
- **Missing Npcap Dependency** — The GNS3 server kept crashing because Npcap, a required packet capture driver, was not installed. The error message "The NPF or NPCAP service is not installed" appeared in the management console. Multiple installation attempts failed with error 0x8007007e before the issue was resolved.
- **Service Conflicts** — Leftover entries from previous failed Npcap installations were blocking new ones. These had to be manually cleared using `sc stop` and `sc delete` commands in an elevated Command Prompt before a clean install could proceed.
- **Learning the GNS3 Interface** — The interface and workflow in GNS3 are quite different from Cisco Packet Tracer. It took time to understand how to add VPCS nodes, connect devices, start nodes, and access their consoles.
- **VPCS Configuration Syntax** — The `ip` command format in VPCS is different from Packet Tracer. Learning the correct syntax and using the `save` command to persist the configuration was a new experience.

All issues were eventually resolved through step-by-step troubleshooting. The experience helped improve problem-solving skills and gave a better understanding of how network simulation tools work at a lower level.

Conclusion

Week 1 of the Network Administration Internship Program at IT-Simplera Institute covered the fundamental concepts of networking and provided hands-on experience with two widely used simulation tools.

The theoretical topics studied during this week — including LAN architecture, IP addressing, network devices, and the OSI and TCP/IP communication models — form the essential base required for network administration work. Building and testing a LAN topology in both Cisco Packet Tracer and GNS3 helped reinforce these concepts through practice.

Cisco Packet Tracer was straightforward to work with for topology design and testing. GNS3 was more complex to set up but offers a simulation environment that is closer to working on real hardware. The challenges faced during the GNS3 setup were a useful learning experience in diagnosing and fixing technical problems.

All objectives for the week were completed. Both topologies were built and configured, all ping tests passed with 0% packet loss, and this report documents the entire process. The knowledge gained this week will be useful for the more advanced topics coming in the following weeks.